Pulkit Katariya

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Education

Indian Institute of Information Technology and Management, Gwalior Gwalior, India

B.Tech in Mathematics and Scientific Computing *CGPA: 8.11/10 | Expected: 2028*

- **Relevant Computer Science Coursework:** Data Structures & Algorithms, Design and Analysis of Algorithms, Object-Oriented Programming, Operating Systems, Software Engineering, Database Systems, Computer Networks, Linear Algebra, Probability & Statistics
- **Certification:** WorldQuant University – Applied Data Science Lab

Technical Skills

Languages: C++, Go, Python, SQL, TypeScript, JavaScript, C, Bash
Systems & Compute: AVX2/SIMD, Cache Tiling, Goto's Algorithm, Linux perf, Memory Optimization, CMake
Backend & Distributed: Distributed Systems, Web Services (SOA), REST APIs, Microservices, FastAPI, Strawberry GraphQL, Concurrency, JWT/RBAC
Databases & Storage: PostgreSQL, Redis (Caching), Apache Parquet, DuckDB, SQLite, Relational Modeling
Cloud & Reliability: Docker, Kubernetes, AWS (EC2, S3), GCP, CI/CD, GitHub Actions, Linux, systemd, Git
Web & Frontend: React, Next.js, Node.js, HTML5, CSS3, Tailwind CSS, Responsive Web Design
Applied ML & Analytics: XGBoost, PyTorch, Scikit-learn, Transformers (DistilBERT), TreeSHAP, MLflow

Open Source Contributions

Volcano – volcano-sh (CNCF Incubating Project)

Open Source Contributor | GitHub

- Authored comprehensive table-driven unit tests for all queue state transitions (`Open`, `Closed`, `Closing`, `Unknown`) in `pkg/controllers/queue/state/`, achieving 100% test coverage across the queue lifecycle for Volcano's HPC and AI batch scheduler.
- Engineered table-driven unit tests for all job controller state transitions in `pkg/controllers/job/state/`, achieving 100% statement coverage integrated into automated CI test suites to prevent regressions in distributed scheduling.

Projects

High-Performance GEMM Engine | GitHub | *C++, AVX2, FMA, Cache Tiling, Linux perf, CMake*

- Engineered a high-performance single-precision General Matrix Multiply (GEMM) kernel in **C++**, progressively optimizing from a naive triple-loop baseline to an AVX2/FMA micro-kernel achieving **70+ GFLOPS**.
- Implemented 3-level cache blocking ($NC \rightarrow KC \rightarrow MC$) and contiguous buffer packing following **Goto's algorithm**, maximizing L1/L2 cache residency and eliminating TLB thrashing.
- Designed a 6×16 register tile utilizing 15 of 16 YMM vector registers to issue 12 FMAs (96 FLOPs) per step, analyzing hardware cache hit rates and throughput using **Linux perf**.

Viral – Distributed ML Pipeline | GitHub | *Go, Python, AWS EC2, Parquet, XGBoost, DistilBERT, FastAPI*

- Architected a 24/7 autonomous **Go ingestion daemon** deployed on **AWS EC2** with systemd supervision; engineered an async HTTP worker pool overcoming a strict **10,000 units/day** API quota across 3,500+ channels at **0.02 units/call**.
- Engineered a 13-checkpoint sampling state machine (0.5h–60h) and comment buffering pipeline, serializing **65,881 trajectories and 4.1M+ comments** into **Snappy-compressed Apache Parquet** partitions.
- Authored 7 ablation studies; built dual-stage comment drift using **DistilBERT** (+21.3% PR-AUC lift) with an optimized **XGBoost** model achieving **0.6206 PR-AUC**, deploying a **FastAPI** microservice delivering sub-2ms predictions.

ACAD-ERP | Live Demo | GitHub | *FastAPI, Strawberry GraphQL, Next.js, React, PostgreSQL, Docker*

- Designed and built a modular academic ERP platform across 9 domains, modeling relational schemas in **PostgreSQL** with strict foreign-key constraints and transactional integrity.
- Implemented an authentication and RBAC system using **JWT HTTPBearer, Argon2 password hashing**, and live role hydration; built fee status tracking with student privacy isolation.
- Architected a unified **Strawberry GraphQL** API layer integrated with a responsive **Next.js / React** web client deployed live on **Vercel** for administrative workflows.